



Case Report

Sudden death due to rupture of the right internal carotid artery in neurofibromatosis type 1: A case report



Yue Liang, Fang Tong, Lin Zhang, Wenhe Li, Yiwu Zhou *

Department of Forensic Medicine, Huazhong University of Science and Technology, Tongji Medical College, No. 13 Hangkong Road, Hankou, Wuhan 430030, PR China

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ABSTRACT

Vascular involvement is a well-recognized manifestation of neurofibromatosis type 1 (NF1) which has the potential to be fatal when disrupted. We here present a case of sudden death due to the fatal arterial rupture resulted from infiltration of the neurofibromas. A 42-year-old man who suffered from NF1 presented a 1-h history of sudden onset of pain in his right cervical region. His condition worsened and became unconscious on his way to the emergency room. Despite resuscitation efforts, he died 30 min later without regaining consciousness. Autopsy examination showed that a neurofibroma located around the right internal carotid artery, confirmed immunohistochemically with S-100, vimentin and CD34. Furthermore, proliferation of spindle cells positive for S-100 was seen in the wall of right internal carotid artery, which was disrupted and resulted in a hemorrhage. These findings suggest that the artery was disrupted by neurofibromas in the vascular wall, which led to fragility of the vessel. On the basis of these findings, we concluded that the cause of death was asphyxia resulting from airway obstruction compressed by the hematoma due to the arterial rupture. As the locality of the neurofibroma and hemorrhage were closed to the carotid baroreflex, we considered another possible mechanism of his sudden death, which could be cardiac inhibition induced by vagal stimulation. We hope this case will increase recognition of NF-1 vasculopathy when encountering any sudden death in NF1 patients.

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1. Introduction

Neurofibromatosis type 1 (NF1), previously known as Von Recklinghausen's neurofibromatosis, one of the most common autosomic-dominant disorders of the nervous system, affects one in 2500–3000 persons worldwide, and shows no gender or ethnic preferences [1]. The responsible gene, located on the chromosome 17q11.2, encodes for the protein neurofibromin and mutation of it result in cell proliferation and inhibited apoptosis throughout the body, which manifests the characteristic neurofibromas [2,3]. In addition, vasculopathy represent another important manifestation which has the potential to be fatal when disrupted [4,5].

Sudden death caused by vasculopathy in NF1 is extremely uncommon and very few cases have been reported in the literature. In this paper, we will present a case of sudden death in a man with NF1 due to the rupture of fragile right internal carotid artery (RICA). An extensive cervical haematoma was formed, causing significant airway compression.

2. Case report

2.1. Clinical data

The decedent, a 42-year-old man, presented with a 1-h history of sudden onset of swelling of the right-side neck with progressive tightness. He experienced severe dyspnea and aggravation of his cervical swelling on his way to the emergency room (ER). Approximately 30 min later, his condition worsened and loss of consciousness subsequently occurred. Several attempts at endotracheal intubation failed as the soft tissues of the neck were swollen and obstructing the larynx. Despite resuscitation efforts, he never regained a cardiac rhythm and he expired. A review of his past medical history was a five-year of multiple neurofibromas and, there was no history of trauma. Suspicious of his death due to malpractice, his relatives asked for medicolegal autopsy to be performed.

2.2. Autopsy findings

Forensic necropsy was performed 48 h after death. Upon external examination, there were no findings of trauma. The most significant finding was limited to the neck, which showed

* Corresponding author.

E-mail addresses: yueyueashin@gmail.com (Y. Liang), fyongfang@gmail.com (F. Tong), lyre2008@126.com (L. Zhang), 18707165089@163.com (W. Li), zhouyiwu@outlook.com (Y. Zhou).

right-sided swelling. We also found asphyxial signs including cyanosis and petechial conjunctival hemorrhages. Freckles could be seen diffusely over the axillary and inguinal region. At least 10 typical café-au-lait spots were observed on the trunk and upper legs, which were flat and brown. A large number of both cutaneous and subcutaneous nodules of different sizes ranging in size from 0.2 to 4.0 cm were unevenly distributed over the entire skin (Fig. 1). The nodules were soft, fleshy, sessile and showed no sign of inflammation. Layer dissection of the neck revealed subcutaneous bleeding at the right side (Fig. 2A), which had spread to the right retropharyngeal wall and the thyrohyal. A large hematoma (10.0 × 7.5 × 6.5 cm) was found adjacent to right carotid bifurcation. Moreover, an oval, off white, fleshy mass measuring 2.5 × 2.0 × 1.3 cm that caused the RICA compressed and displaced was located within this perivascular hematoma (Fig. 2B). The RICA had a tiny defect in its extracranial segment adjacent to the mass, approximately 2 cm distal to the carotid bifurcation, which was the predominant source of the hemorrhage. No aneurysmal dilatation of the artery was noted. Edema of the retropharyngeal wall was remarkable. The swelling within the larynx had caused significant obstruction of the airway and was associated with mucosal edema that involved the epiglottis, especially on the left side. The right side of the esophageal adventitia showed sheet hemorrhage.

Additional internal examination findings included cerebral edema and sporadic hemorrhagic spots of the lungs. The subsequent toxicology screening was negative.

2.3. Histological findings

The mass surrounding the wall of RICA revealed an irregular pattern of proliferative elongated spindle cells with waxy or comma-shaped nuclei embedded in a stroma composed of wire-like collagen fibers and amorphous mucoid substances. There was no evidence of malignancy, such as cellularity, cellular pleomorphism, or mitoses. Immunohistochemistry (IHC) of the mass revealed positivity for S-100 protein of the nuclei and cytoplasm of the spindle cells and most cells within the fibrous connective tissue were vimentin and CD34 positive.

The rupture site was identified on consecutive sections of RICA near the mass both in hematoxylin-eosin (HE) and Elastica-Masson's stains, which demonstrated focal loss of the internal elastic lamella and the media, and fragmentation or thinning of the adventitia with hemorrhage and collagen deposition in area of the adventitia. Concomitantly, a same lesion which showed diffuse proliferation of spindle cells with waxy nuclei and a mucoid change of the stroma widely infiltrated the wall of RICA. The lesion was S-100 positive, confirming the same origin with the

surrounding mass (Fig. 3). On the other hand, a reduction of elastic fibers and focal loss of the muscularis were observed in the media near the disrupted site. And increased collagen deposition in areas of muscularis loss that is best appreciated with Elastica-Masson's stain, is observed (Fig. 4).

Histological changes of cutaneous nodules were consistent with the mass, which was positive for S-100 protein. Examination of all other organs was unremarkable except for sporadic eosinophilic degeneration of neuronal cells in brain.

3. Discussion

NF1 is an autosomal dominant condition caused by heterozygous mutations of the NF1 gene. Most common characteristics include neurofibromas, café-au-lait spots, lisch nodules and axillary freckling. The neurofibromas, the hallmark feature of NF1, are benign tumors consisting of Schwann cells, perineurial cells, fibroblasts, mast cells, axons, and blood vessels and as well large amounts of intercellular material, including collagen and amorphous ground substances, for example, mucopolysaccharides and glycosaminoglycans [2,6,7]. They are positive for S-100 protein at IHC analysis, indicating their neural origin from Schwann cells [8]. Furthermore, positivity for vimentin and CD34 are also indicative of neurofibromas [9,10]. In this case, neurofibromas were demonstrated by macroscopic, histopathological and immunohistochemical characteristics. Consequently, according to the clinical diagnostic criteria [11], the patient who presented with more than six café-au-lait spots, axillary and inguinal freckling, multiple cutaneous and subcutaneous neurofibromas, was diagnosed with NF1.

Mortality associated with NF1 is about 0.92 per 1 million people and approximately a 15-year decrease in life expectancy of NF1 patients has been revealed in some studies. Fatal complications of NF1 resulting in death include malignant peripheral nerve sheath tumor, CNS tumors, spinal cord compression by neurofibromas, organ compression by neurofibromas and pheochromocytomas [12]. In addition, vasculopathy has been reported as an uncommon complication which is another cause of death in NF1 patients [4,5].

NF1 vasculopathy was first described in 1945 [13]. The prevalence is about 0.4% to 6.4% and it is often asymptomatic, though it has a potential for significant morbidity and mortality [14]. There are two distinct pathogenetic mechanisms of vasculopathy: smooth muscle dysplasia and direct vascular invasion by neurofibromatous tissue. Lesions manifest as stenosis, dissection, aneurysm formation, or rupture of the affected vessels [15]. Histopathological characteristics of the affected vessels are (1) surrounded by neurofibromatous tissue, (2) abnormal proliferation of spindle or epithelioid cells disturbing the integrity of the vessel, (3) thinning of the media and fragmentation of the muscularis and elastic layer, (4) intimal thickening with occlusion of the vascular lumen [16–18]. It is suggested that proliferation of spindle cells occurred within arterial walls leads to dysplasia of smooth muscle layers and destruction of the media and the elastic fibers. These pathological changes may increase fragility of the vessels gradually and are thought to be responsible for the associated arterial disruption [19]. In our case, this explanation is reasonable. In addition to the absent of intimal thickening, the most striking histological changes was the disintegration of the vascular structure by proliferation of spindle cells which were positively stained for S-100, suggesting their neural origin. On the other hand, other than a proliferation within the vessel microscopically**, a macroscopic neurofibroma was formed adjacent to the artery. These findings represent that the onset of vascular lesion is mainly due to direct tumor invasion which caused the arterial extreme fragility. Therefore, it is at increased risk of rupture which was spontaneous or



Fig. 1. A large number of nodules unevenly distributed over the skin, café-au-lait spots (arrow) were observed on the trunk and extremities.

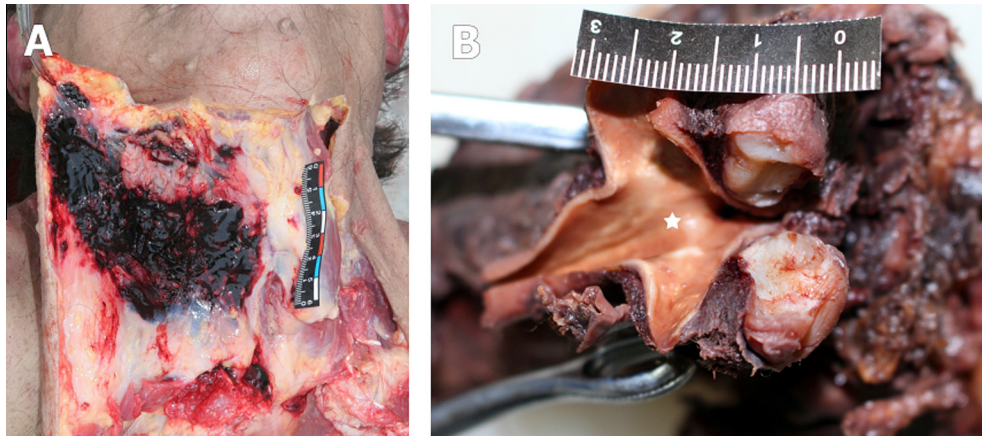


Fig. 2. (A) Subcutaneous bleeding presented at the right cervical region. (B) A large hematoma was found adjacent to carotid bifurcation (asterisk) and there was an oval, off white, fleshy mass inside of it (the cut section).

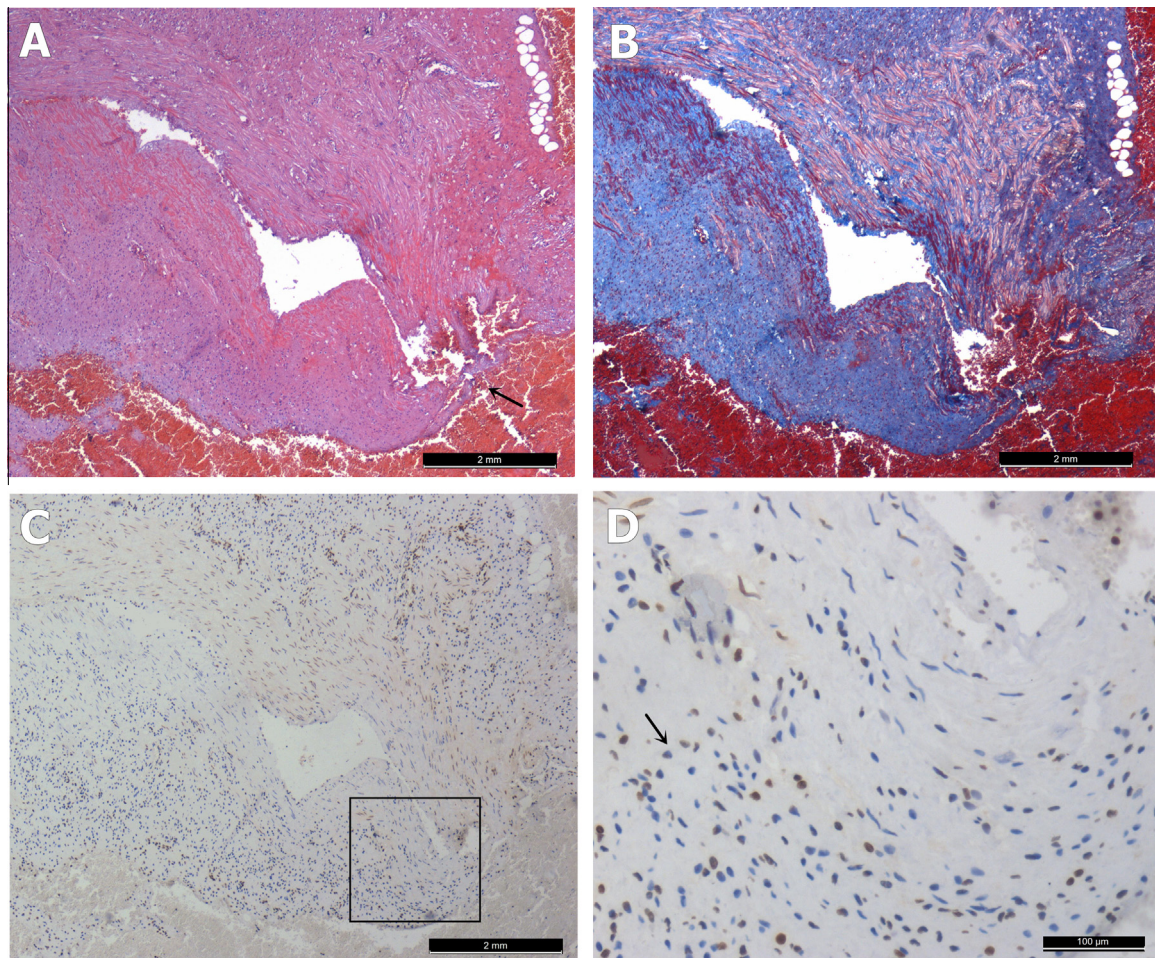


Fig. 3. (A and B) The wall of RCA stained with both in HE (A) and Elastica–Masson's (B) stains. The rupture site (arrow in panel A) showed focal loss of the internal elastic lamella and the media, and fragmentation or thinning of the adventitia with hemorrhage and collagen deposition in area of the adventitia (A, HE stain, 40 $\times$; (B) Elastica–Masson's stain, 40 $\times$). (C) Infiltration by spindle cells was seen in the wall of RCA, which expressed S-100 protein (IHC stain, 40 $\times$). (D) S-100 immunostaining (square in panel C). The arrow indicates the adventitia (IHC stain, 200 $\times$).

secondary to trauma. However, we couldn't find any evidence of trauma in autopsy and there was no history of trauma. We can exclude the possible of traumatic rupture of RCA.

Thus, we can conclude that the artery terminated with rupture as a result of compressive force from surrounding neurofibroma-

tous tissue on the basis of the arterial extreme fragility. And excess movement of his neck without a special awareness might be another possible trigger of the rupture. Combined with the clinical data, we know that the patient had gradual swelling of the neck before death, which were initiated by rapid expansion of cervical

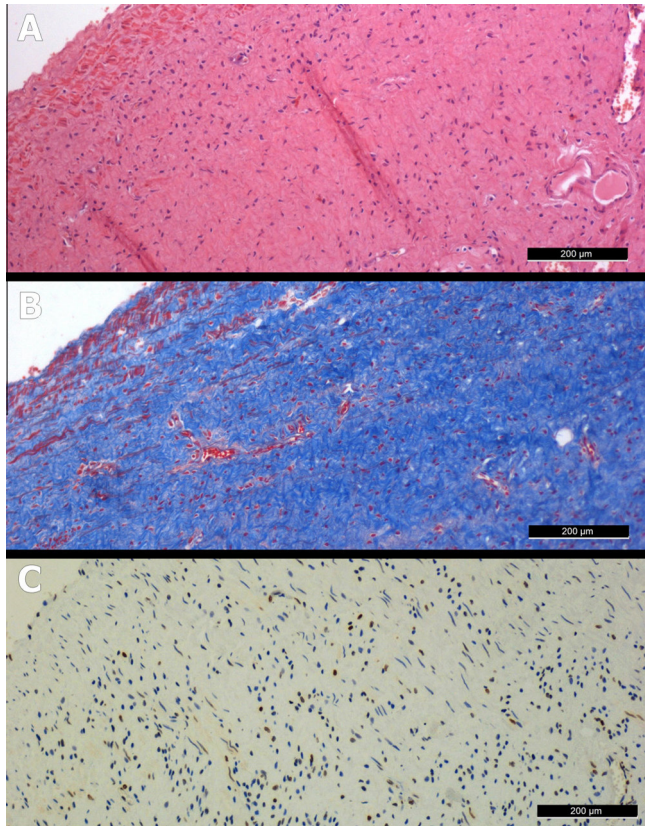


Fig. 4. Wall of the RICA. (A and B) The arterial wall showed a reduction of elastic fibers in the media, with a few residual smooth muscle fibers, using both HE and Elastica–Masson's stains (A, HE stain, 100 $\times$; B, Elastica–Masson's stain, 100 $\times$). (C) Infiltration by spindle cells in the artery wall, which expressed S-100 protein (IHC stain, 100 $\times$).

subcutaneous bleeding caused by the arterial rupture, and that this accounted for his presenting symptoms of neck tightness and dyspnoea. Due to the underlying anatomy of the neck, bleeding spread to the cervical region through the submandibular space and resulted in the formation of a cervical subcutaneous hematoma. A critical point was suddenly reached when the hematoma swelling caused complete airway obstruction and rapid deterioration. Consequently, according to the macroscopic and histological findings, we concluded that the cause of death was asphyxia resulting from airway obstruction caused by compression of the hematoma due to rupture of the artery and the severe edema of the larynx. In addition, it is generally accepted that a blow or a compression of the neck stimulating the carotid sinus can lead to circulatory collapse resulting from activation of an arterial baroreflex controlling the heart rate [20]. In the case herein described, as the locality of the neurofibroma and hemorrhage were closed to the carotid baroreflex, considering how quickly loss of consciousness occurred, we considered another possible mechanism of his sudden death could be cardiac inhibition induced by vagal stimulation.

NF1 vasculopathy has been observed most often in the renal arteries with 41% occurrence, but no vessel is immune. Other involvement sites include the neck and head (19%), extremities (12.9%), and the abdominal aorta (12%) [14]. Several case reports concerning spontaneous rupture of arteries in NF1 have been reported, generally describing vascular stenosis, increased fragility, aneurysm formation, and spontaneous rupture [21]. However, a fatal outcome may be avoided with early reorganization and hemostatic treatment. Thus a fatal hemorrhage due to rupture of a major artery is extremely uncommon. In 2008, Hinsch et al. [22] reported a young NF1 woman with lethal bleeding due to

spontaneous rupture of a lumbar and renal artery and died during the surgery. The postmortem examination revealed high fragility of several arteries and veins in the abdominal resulted in a destroyed vessel structure caused by invading neurogen fibers, which explained their spontaneous rupture, and the impossibility to suture the blood vessels. Spontaneous haemothorax, one more dramatic presentation of vasculopathy, has been reported as a cause of sudden death [23]. In both two cases, death was attributed to hypovolemic shock resulting from massive bleeding which could be excluded in our case as the bleeding was not so serious as to be a cause of death and no sign pointed to secondary changes related to massive bleeding.

As illustrated in this case, vasculopathy in NF1 which may lead to increased vascular fragility can be life threatening, particularly in the setting of major arterial rupture. Such cases are relatively rare in both clinical and forensic medicine, the forensic pathologists should recognize this possible complication when encountering any sudden death in NF1 patients. Symptoms before death such as neck tightness and dyspnoea, review of the medical records and the histopathology examination, IHC stain in particular, serve a guiding role in determination of the cause of death.

Conflict of interest

None.

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